

WASP-43b

R-0503

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_230228 · created 2026-07-20T17:03:38+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460003.78848 +/- 0.00099 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1981 +/- 0.0078
Transit depth (Rp/Rs)^2	3.92 +/- 0.31 [%]
Orbital Inclination (inc)	80.51 +/- 0.46
Airmass coefficient 1 (a1)	0.9956 +/- 0.0081
Airmass coefficient 2 (a2)	0.003 +/- 0.018
Scatter in the residuals of the lightcurve fit is	0.81 %
Best Comparison Star	#2 - [575, 224]
Optimal Aperture	3.56
Optimal Annulus	8.64
Transit Duration (day)	0.0482 +/- 0.0019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1981 $\pm$ 0.0078 vs 0.1594 (deviation 4.62 σ)
Duration fit vs published	0.0482 d vs 0.0512 d (deviation 1.49 σ)
Mid-time O-C	-1.5 min
Depth SNR	23.6
Residual scatter	0.81 %

Light curve

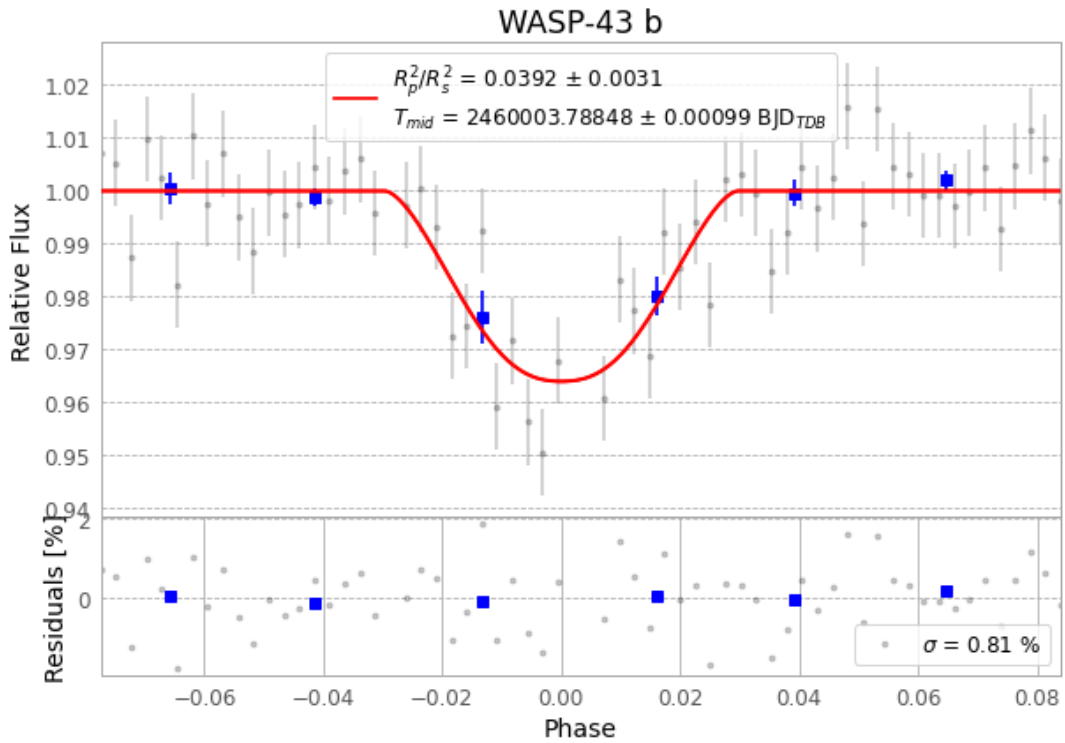


Figure 1 — FinalLightCurve_WASP-43 b_2023-02-27.png (SHA-256 cb1ac20dc9de094e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [453, 136], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8156bd6bfa3a63c4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

64 FITS frames (42 MB), WASP-43230228051515.FITS → WASP-43230228082415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-230228.log (d9b08cf83189...), FinalLightCurve_WASP-43 b_2023-02-27.png (cb1ac20dc9de...), FinalLightCurve_WASP-43 b_2023-02-27.csv (b9bd0d928abc...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 17:03Z.